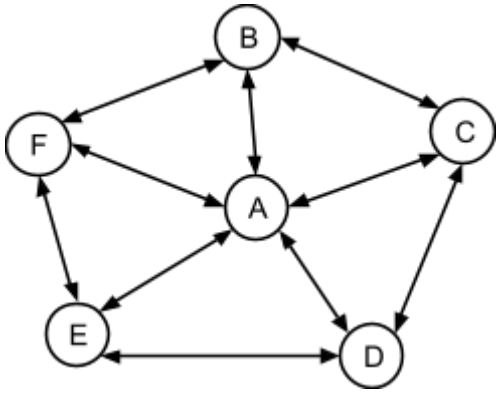
American Computer Science League

2024-2025 ● Contest 4: Short Problems ● Classroom Division

1. Graph Theory

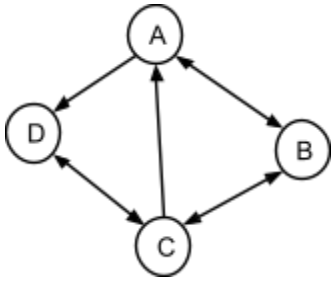
How many 1s are in the adjacency matrix for the following graph?



A. 10
B. 15
C. 20
D. 30
E. 36

2. Graph Theory

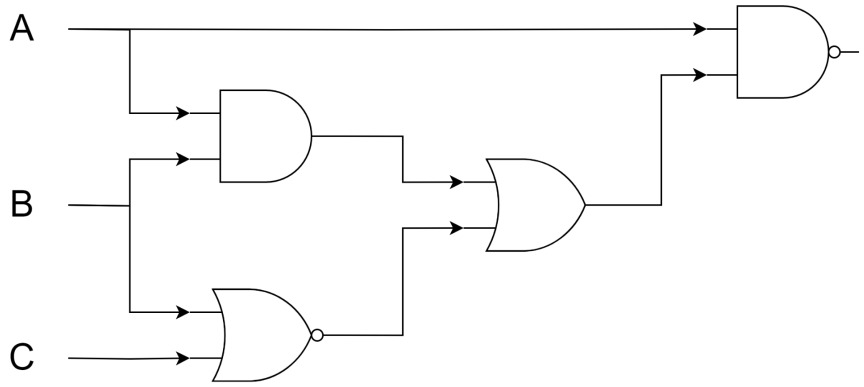
How many cycles are there in the directed graph below?



A. 3
B. 4
C. 5
D. 6
E. 7

3. Digital Electronics

Translate the following digital circuit into a Boolean expression.



A. $\overline{A\,(AB\,+\overline{B\,+\,C})}$

B. $\overline{A\,(A\,+\,B\,+\overline{BC})}$

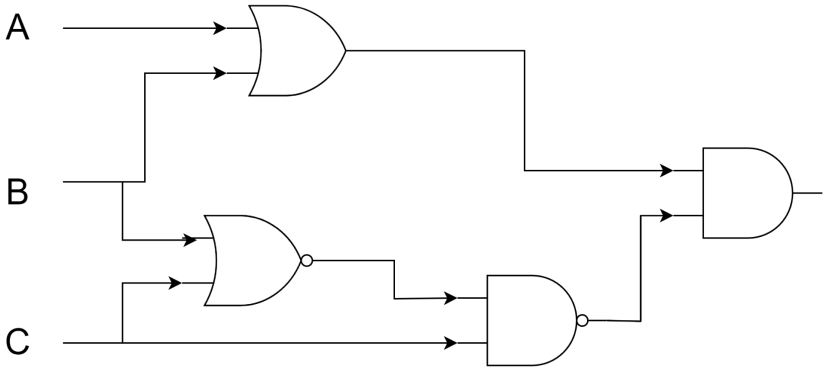
C. $\overline{\overline{A}\,(AB\,+\,\overline{B\,+\,C})}$

D. $\overline{A\,+\,AB\,+\overline{BC}}$

E. $\overline{A\,(\overline{AB}\,+\overline{B\,+\,C})}$

4. Digital Electronics

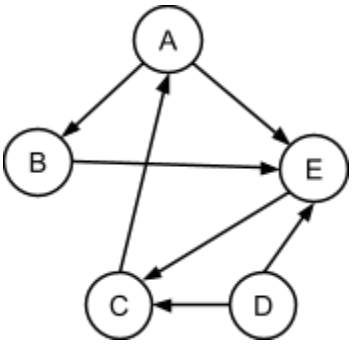
How many ordered triples make the following digital circuit TRUE?



A. 2
B. 3
C. 4
D. 5
E. 6

5. Graph Theory

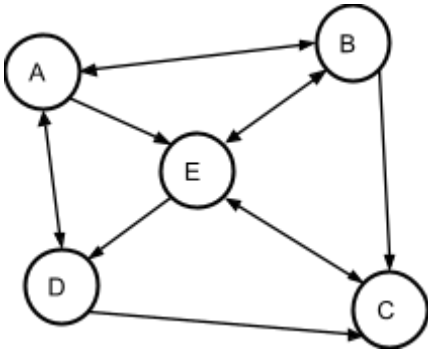
How many paths of length 2 are there in the following directed graph?



A. 6
B. 7
C. 8
D. 9
E. 10

6. Graph Theory

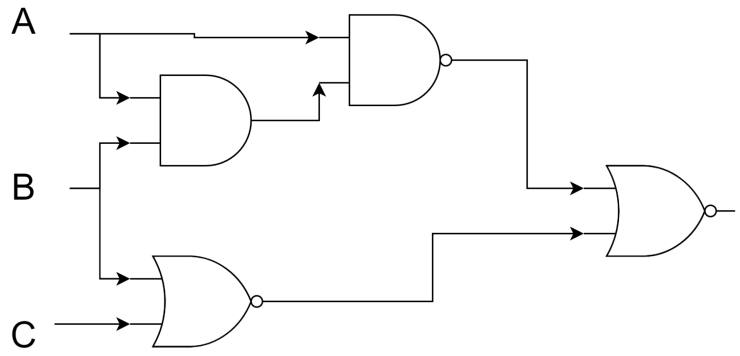
How many cycles are there in this directed graph?



A. 8
B. 9
C. 10
D. 11
E. 12

7. Digital Electronics

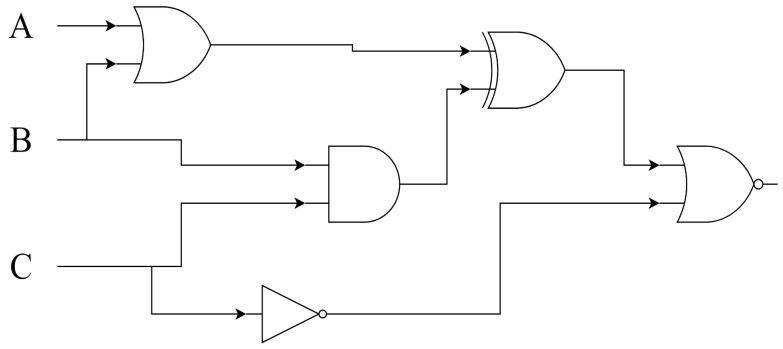
How many ordered triples make the following digital circuit TRUE?



A. 0
B. 2
C. 4
D. 6
E. 8

8. Digital Electronics

Translate the following digital circuit to a Boolean algebra expression.



A. $\overline{AB \oplus (B + C) + \overline{\overline{C}}}$
B. $\overline{(A + B) \oplus \overline{BC}} + \overline{C}$
C. $\overline{A + B \oplus BC + \overline{\overline{C}}}$
D. $\overline{(A + B) \oplus BC + C}$
E. $\overline{(A + B) \oplus BC + \overline{\overline{C}}}$

9. Assembly Language

What is the output when the following program is executed?

```
NUM     DC      2025
        LOAD    NUM
        DIV     =100
        STORE   A
        MULT    =100
        SUB     NUM
        MULT    =-1
        STORE   B
        DIV     =5
        MULT    =10
        STORE   ANS
        LOAD    A
        DIV     =5
        ADD     ANS
        STORE   ANS
        PRINT   ANS
        END
```

A. 45
B. 54
C. 405
D. 504
E. 99

10. Assembly Language

What is the output when the following program is executed?

```
T       DC      0
N       DC      19
NUM     DC      135
TOP     LOAD    NUM
        SUB     T
        BE      STOP
        LOAD    N
        ADD     =1
        STORE   N
        LOAD    T
        ADD     N
        STORE   T
        BU      TOP
STOP    PRINT   N
        END
```

A. 21
B. 22
C. 23
D. 24
E. 25